Omni-Compass benchmark: Kubernetes native vs Kubernetes + Omni-Compass

Two identical kind clusters (1 control plane + 6 workers), same add-ons, workload (php-apache + HPA, target 50), load schedule and capture. **Native**: Omni-Compass not running. **Omni**: Omni-Compass drives the HPA target, the node pool (cordon/drain/uncordon), the power cap (in-place CPU limits on the app pods, enforced by the kernel), heat (harness law on live power), security (hold signal) and the rollout guard. Response times are real HTTP requests timed every 5 s. Power is a declared model (idle 100 W + 150 W x utilisation per worker in service; parked workers on standby at idle power), not a meter.

Gauges

Gauge	Native	Omni-Compass	Change
duration (min)	20.15	20.13	
worker nodes in service, mean	6.00	6.00	+0.0%
worker nodes in service, min	6.00	6.00	+0.0%
worker nodes in service, max	6.00	6.00	+0.0%
node-hours	2.01	2.01	-0.1%
power (W), mean	628	643	+2.3%
power (W), peak	651	681	+4.6%
energy (Wh)	211	216	+2.2%
energy, parked workers still on at idle power (Wh)	211	216	+2.2%
CPU used (cores), mean	0.872	1.26	+44.4%
CPU allocatable (cores), mean	24.00	24.00	+0.0%
utilisation (used / allocatable)	0.036	0.052	+44.4%
energy per core-hour (Wh)	721	510	-29.2%
node-hours per core-hour	6.88	4.76	-30.8%
pending pods, pod-minutes	0.267	0.000	-100.0%
pending pods, peak	1.00	0.000	-100.0%
HPA replicas, mean	8.95	9.48	+5.9%
HPA replicas, peak	10.00	10.00	+0.0%
HPA shortfall (desired > current), minutes	0.267	0.800	+200.0%
requests timed	1,055	1,095	+3.8%
time over the response line (% of samples)	0.853	0.000	-100.0%
response time (ms), mean	139	97.53	-29.7%
response time (ms), median	116	85.40	-26.5%
response time (ms), 95th percentile	293	152	-48.1%
response time (ms), 99th percentile	476	171	-64.0%
failed requests (%)	0.000	0.000	0

Significance (2-minute blocks: native 10, omni 9; bootstrap 95% interval)

Metric	Native	Omni-Compass	Change	95% CI of difference	Verdict
node_hours_per_core_hour	7.4962	5.2103	-30.5%	[-3.9286, -0.6338]	GAIN
kwh_per_core_hour	0.7818	0.5548	-29.0%	[-0.3949, -0.0691]	GAIN
utilisation	0.0359	0.0534	+48.8%	[+0.0047, +0.0316]	GAIN
pending_pod_minutes_per_hour	0.7619	0.0000	-100.0%	[-2.2857, +0.0000]	STEADY, no significant difference
hpa_shortfall_minutes_per_hour	0.7742	2.4932	+222.0%	[-1.5484, +5.0914]	STEADY, no significant difference

What Omni-Compass did

- decisions: 120 (one per minute)
- HPA target writes: 0
- node-pool resizes: 3
- scheduling-floor adds (pending pods): 0
- power-cap pod resizes (in place): 0
- rollout actions: 4
- power cap decided per minute: 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 0.695 0.695 0.695 1.0 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 1.0 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 1.0 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695
- heat (thermal) per minute: 0.327 0.333 0.34 0.346 0.353 0.359 0.365 0.371 0.376 0.382 0.387 0.393 0.398 0.403 0.408 0.412 0.417 0.421 0.426 0.43 0.434 0.438 0.443 0.447 0.451 0.455 0.458 0.462 0.466 0.469 0.473 0.476 0.479 0.482 0.485 0.488 0.491 0.494 0.497 0.5 0.503 0.505 0.508 0.511 0.513 0.516 0.519 0.521 0.523 0.526 0.528 0.53 0.533 0.535 0.537 0.539 0.541 0.543 0.545 0.547 0.548 0.55 0.552 0.553 0.555 0.556 0.558 0.559 0.56 0.56 0.561 0.562 0.563 0.564 0.565 0.566 0.567 0.567 0.568 0.569 0.57 0.57 0.571 0.572 0.573 0.574 0.575 0.576 0.576 0.577 0.578 0.579 0.58 0.58 0.581 0.582 0.582 0.583 0.584 0.584 0.585 0.585 0.586 0.586 0.587 0.587 0.588 0.588 0.588 0.588 0.589 0.589 0.589 0.589 0.589 0.59 0.59 0.59 0.59 0.591
- nodes decided per minute: 6 6 6 6 6 6 6 6 6 6 6 6 6 6 6 6 6 5 5 5 5 5 5 5 5 5 4 4 4 4 4 4 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
- HPA target decided per minute (%): 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77

Reset

```
restored target: 50
workers in service: 6 of 6
Omni records left after kill: none
```

Minute by minute

min	nodes native	nodes omni	power W native	power W omni	replicas native	replicas omni	pending native	pending omni	CPU native	CPU omni
0	6.00	6.00	615	621	5.00	5.00	0.000	0.000	0.509	0.681
1	6.00	6.00	617	628	5.00	7.00	0.000	0.000	0.533	0.860
2	6.00	6.00	617	624	5.00	9.00	0.000	0.000	0.590	0.762
3	6.00	6.00	624	634	5.75	9.00	0.000	0.000	0.755	1.02
4	6.00	6.00	635	642	9.00	10.00	0.000	0.000	1.03	1.35
5	6.00	6.00	630	646	10.00	10.00	0.000	0.000	0.922	1.34
6	6.00	6.00	632	649	10.00	10.00	0.000	0.000	0.970	1.43
7	6.00	6.00	645	674	10.00	10.00	0.000	0.000	1.31	2.12
8	6.00	6.00	645	678	10.00	10.00	0.000	0.000	1.32	2.15
9	6.00	6.00	646	676	10.00	10.00	0.000	0.000	1.33	2.17
10	6.00	6.00	644	669	10.00	10.00	0.000	0.000	1.30	1.94
11	6.00	6.00	618	625	10.00	10.00	0.000	0.000	0.627	0.786
12	6.00	6.00	616	624	10.00	10.00	0.000	0.000	0.562	0.807
13	6.00	6.00	625	632	10.00	10.00	0.250	0.000	0.775	0.970
14	6.00	6.00	631	650	10.00	10.00	0.000	0.000	0.957	1.45
15	6.00	6.00	633	654	10.00	10.00	0.000	0.000	0.983	1.58
16	6.00	6.00	632	648	10.00	10.00	0.000	0.000	0.966	1.31
17	6.00	6.00	630	631	10.00	10.00	0.000	0.000	0.884	0.945
18	6.00	6.00	620	623	10.00	10.00	0.000	0.000	0.653	0.745
19	6.00	6.00	618	626	10.00	10.00	0.000	0.000	0.592	0.792

Limits

- kind nodes are containers sharing one CI machine; timings and CPU are noisier than real servers.
- The two arms ran on two different CI machines at the same time; machine-to-machine variation is part of the noise.
- Energy is modelled from utilisation and nodes in service with the declared constants above.